

MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES
(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting – 6_Lab_D1_AWK

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AWK Pattern Block Questions

For every question:

Execute the command using employee.txt.

Part A Numerical and String Comparisons

Display the name, department, and salary of employees earning more than 70000.
Display employees whose salary is between 60000 and 80000, including both limits.
Display employees whose employment status is Inactive.
Display employees belonging to the VLSI department.

```
awk '$3 > 70000 {print $1, $2, $3}' employee.txt
awk '$3 >= 60000 && $3 <= 80000 {print $1, $2, $3}' employee.txt
awk '$4 == "Inactive" {print $1, $2, $4}' employee.txt
awk '$2 == "VLSI" {print $1, $2, $3}' employee.txt
```

Part B Regular-Expression Patterns

Display all records containing the word Security.
Display employees whose names begin with the letters A, B, or C.
Apply the regular expression only to the employee-name field.
Display employees whose department field does not match Security.

```
awk '/Security/ {print}' employee.txt
awk '$1 ~ /^[ABC]/ {print}' employee.txt
awk '$1 ~ /^[ABC]/ {print}' employee.txt
awk '$2 !~ /Security/ {print}' employee.txt
```

Part C Logical Patterns

Display employees who work in the Security department and earn more than 70000.
Display employees who belong to either the VLSI or Networking department.
Display employees who:

- **Work in the VLSI department**
- **Earn more than 70000**
- **Have Active status**

```
awk '$2 == "Security" && $3 > 70000 {print}' employee.txt
awk '$2 == "VLSI" || $2 == "Networking" {print}' employee.txt
awk '$2 == "VLSI" && $3 > 70000 && $4 == "Active" {print}' employee.txt
```

Part D Conditional Pattern

Department-dependent salary condition

Apply different salary conditions:

If the employee belongs to Security, select employees earning more than 75000. Otherwise, select employees earning more than 70000.

Display the employee name, department, and salary.

```
awk '($2 == "Security" && $3 > 75000) || ($2 != "Security" && $3 > 70000) {print
$1, $2, $3}' employee.txt
```

Part E Grouped Patterns

Display active employees who belong to either the Security or VLSI department.

The Active condition must apply to both departments.

```
awk '$4 == "Active" && ($2 == "Security" || $2 == "VLSI") {print}' employee.txt
```

Display employees who satisfy one of these conditions:

Work in Security and earn more than 80000
Work in VLSI and earn more than 75000

```
awk '($2 == "Security" && $3 > 80000) || ($2 == "VLSI" && $3 > 75000) {print}'
employee.txt
```

Display active employees who satisfy one of these conditions:

Work in Security and earn more than 70000
Work in VLSI and earn more than 75000

```
awk '$4 == "Active" && (($2 == "Security" && $3 > 70000) || ($2 == "VLSI" && $3 > 75000)) {print}' employee.txt
```

Part F Negated Patterns

Display employees whose status is not Active.
Use an exact field comparison because the word Inactive contains the text Active.
Display employees who do not belong to either the Security or VLSI department.

```
awk '$4 != "Active" {print}' employee.txt
awk '$2 != "Security" && $2 != "VLSI" {print}' employee.txt
```

Part G Range Patterns

Display all records beginning with employee ID 103 and ending with employee ID 108. Include both boundary records.

```
awk '$1=="103", $1=="108" {print}' employee.txt
```

Display all records beginning with David and ending with Irene. Include both boundary records.

```
awk '$1=="David", $1=="Irene" {print}' employee.txt
```

Part H Built-In Variables

Display only the first five records. Include the record number in the output.

```
awk 'NR <= 5 {print NR, $0}' employee.txt
```

Display records numbered 4 through 9 using the built-in variable NR.

```
awk 'NR >= 4 && NR <= 9 {print}' employee.txt
```

Display only records containing exactly five fields. Print the record number and the complete record.

```
awk 'NF == 5 {print NR, $0}' employee.txt
```

Display only odd-numbered records using NR.

```
awk 'NR % 2 == 1 {print NR, $0}' employee.txt
```

Part I Arithmetic Conditions

Assume each employee receives a bonus equal to 10% of the salary. Display employees whose calculated bonus exceeds 7000.

Print:

Employee name
Salary
Calculated bonus
Question 24 Revised salary

```
awk '($3 * 0.10) > 7000 {print "Employee name:", $1, "Salary:", $3, "Calculated
bonus:", ($3 * 0.10), "Question 24 Revised salary:", ($3 + ($3 * 0.10))}'
employee.txt
```

Calculate the revised salary after adding a 12% increment. Display employees whose revised salary exceeds 85000.

```
awk '($3 * 1.12) > 85000 {print $1, $2, $3, ($3 * 1.12)}' employee.txt
```

Calculate the monthly salary by dividing the annual salary by 12. Display employees whose monthly salary exceeds 6000.

```
awk '($3 / 12) > 6000 {print $1, $2, $3, ($3 / 12)}' employee.txt
```

Part J Integrated Challenge

Generate a report containing active employees who satisfy one of these conditions:

Work in Security and earn at least 70000
Work in VLSI and earn at least 75000
Work in Networking and earn at least 70000

```
awk '$4 == "Active" && (($2 == "Security" && $3 >= 70000) || ($2 == "VLSI" && $3 >=
75000) || ($2 == "Networking" && $3 >= 70000)) {print}' employee.txt
```

The output must contain:

Employee ID

Employee name
Department
Salary
Status

```
awk '$4 == "Active" && (($2 == "Security" && $3 >= 70000) || ($2 == "VLSI" && $3 >=
75000) || ($2 == "Networking" && $3 >= 70000)) {print $1, $2, $3, $4}' employee.txt
```

Add a report heading using BEGIN and print the total number of selected employees using END.

```
awk '
BEGIN {
  print "Employee Report"
  print "ID\tName\tDepartment\tSalary\tStatus"
}
$4 == "Active" && (($2 == "Security" && $3 >= 70000) || ($2 == "VLSI" && $3 >=
75000) || ($2 == "Networking" && $3 >= 70000)) {
  print $1, $2, $3, $4
  count++
}
END {
  print "Total selected employees:", count
}
' employee.txt
```